

Capsinoids and related food ingredients activating brown fat thermogenesis and reducing body fat in humans.

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PURPOSE OF REVIEW: Capsaicin and its nonpungent analog (capsinoids) are known to be food ingredients that increase energy expenditure and decrease body fat. This article reviews the role of brown adipose tissue (BAT) for the thermogenic effect of these compounds in humans and proposes the possibility of some other antiobesity food ingredients. RECENT FINDINGS: A single oral ingestion of capsinoids increases energy expenditure in human individuals with metabolically active BAT, but not those without it, indicating that capsinoids activate BAT and thereby increase energy expenditure. This finding gave a rational explanation for discrepant results of the effects of capsinoids in the previous studies. Human BAT may be largely composed of inducible 'beige' adipocytes more than typical brown adipocytes because its gene expression patterns are similar to beige cells isolated from murine white fat depots. In fact, preadipocytes isolated from supraclavicular fat deposits - where BAT is often detected - are capable of differentiating into brown-like adipocytes in vitro, providing evidence of inducible brown adipogenesis in adult humans. SUMMARY: As human BAT may be inducible, a prolonged ingestion of capsinoids would recruit active BAT and thereby increase energy expenditure and decrease body fat. In addition to capsinoids, there are numerous food ingredients that are expected to activate BAT and so be useful for the prevention of obesity in daily life.